Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign GPA: 3.95/4.0

Master of Computer Science (MCS), Planned *August 2027 – May 2028*

Bachelor of Science in Computer Science & Geographic Information Science *August 2024 – May 2027*

- Involvement: ACM SIG Project Lead for HCI and AI, Virtual Reality Physics Simulation Developer
- Relevant Coursework: Data Structures, Algorithms & Models of Computation, Computer Systems, Database Management in the Cloud, Applied Linear Algebra, Spatial Analysis, Virtual Reality, Human-LLM Interaction

Experience

UIUC CyberGIS Lab Urbana, IL

Machine Learning Engineering Intern *September 2026 – Present*

- Benchmarking the lab's geospatial AI agent on all 50 GeoAnalystBench tasks, showing it matches published LLMs on code but plans more steps than the expert on 43 of 48 tasks
- Uncovering 4 bugs that break core features of the live agent by user-testing dataset search, map projections, raster data, file uploads, and adversarial prompts
- Reworking the agent's tool selection, search indexing, and code execution to target over-planning and bugs from testing

Semantic Segmentation Failure Analysis Urbana, IL

Undergraduate Researcher | shanmukhupad.github.io/portfolio/seg.pdf *May 2026 – August 2026*

- Trained a DeepLabV3-ResNet50 model on aerial imagery and built a tiled scanner that maps where it is likely wrong across full-size images, using only the model's own confidence and no labeled data
- Validated the error maps on 4 benchmark datasets, found the model is about half as accurate along object edges, and presented the work in an IEEE-format paper and a CyberGIS talk

ACM SIG-CHI | github.com/sigchi-uiuc/digital-diary Urbana, IL

Project Lead *September 2025 – April 2026*

- Engineered a real-time camera pipeline with OpenCV and MediaPipe that tracks face and hand landmarks from webcam video to detect mood and gesture controls, running as its own Docker service
- Led a 15-person team building a Next.js journaling app with a guided mode that pulls in Google Calendar, weather data, and Claude AI

UIC Electronic Visualization Laboratory | github.com/ShanmukhUpad/curio Chicago, IL

Software Engineering Intern *June 2025 – August 2025*

- Developed a Pandas merge tool that cut processing time by about 30% across hundreds of Chicago urban data pipelines
- Added dataset merging and multi-tab data views to Curio, a visual data-pipeline tool used by researchers

Projects

Multi-Agent LLM Sycophancy | shanmukhupad.github.io/portfolio/mas.pdf March 2026 – May 2026

- Demonstrated that one injected false claim causes a 100% cascade through hierarchical LLM agentic workflows by turn 3, measured without an LLM judge using stance flips, planning errors, and deference language
- Found that anti-sycophancy prompting does not stop the collapse, and co-authored an ACL-format paper on it

Off the Clock: Standalone VR Game | *C#, Unity XR* | *Virtual Reality Course Project* April 2026 – May 2026

- Built motion controls and a combat system for sword fighting, laser guns, and mech flight, using controller tracking data and sphere-cast sweeps so fast sword swings register hits reliably
- Held Meta Quest 3 builds to a 90 FPS target with an in-headset profiling overlay, and fixed laser effects that vanished in the headset by replacing shaders that lacked single-pass stereo support

Technical Skills

Languages: Python, C++, C, C#, Java, TypeScript, SQL, R
Computer Vision & Image Processing: OpenCV, PyTorch, torchvision, MediaPipe, NumPy, ModernGL/GLSL
Tools: Git, Docker, Android, Unity XR, React, Next.js, XGBoost, GeoPandas, ArcGIS Pro